

Credit Risk Intelligence Report

LR | Naive Bayes | Random Forest | XGBoost + SHAP

1. Dataset Summary

Metric	Value
Total Clients	26,177
Default Rate	33.01%
Median Credit Limit	\$50,000
Mean Age	47.5 years
Engineered Features	3 (CREDIT_UTILIZATION, REPAYMENT_RATIO, CONSECUTIVE_DELAYS)
Models Evaluated	4 (LR, Naive Bayes, Random Forest, XGBoost)

2. Model Performance Comparison (Same Test Set)

Model	AUC-ROC	Strength	Limitation
Logistic Regression	0.6814	Interpretable, coefficients	Linear boundary only
Naive Bayes (tuned)	0.6241	Fast, probabilistic output	Feature independence assumption
Random Forest (tuned)	0.7201	Captures non-linearity	No direct probability calibration
XGBoost ★	0.7210	Best AUC + SHAP explainability	Requires careful tuning

3. Top Predictive Features — XGBoost via SHAP

Rank	Feature	Mean SHAP	Financial Meaning
1	PAY_0	0.5845	Most recent payment delay — strongest signal
2	PAY_2	0.3708	Payment delay 2 months ago
3	PAY_3	0.2156	—
4	PAY_AMT1	0.2083	—
5	BILL_AMT1	0.1468	Most recent bill amount
6	BILL_AMT3	0.0764	—
7	LIMIT_BAL	0.0588	Credit limit — bank creditworthiness proxy
8	AGE	0.0565	Client age — financial experience proxy

4. Hypotheses Verdict

ID	Hypothesis	Verdict
H1	PAY_0 delay → default risk	✓ Confirmed — PAY_0 is top feature in RF & XGBoost (SHAP)
H2	High credit utilization → risk	✓ Confirmed — CREDIT_UTILIZATION separates classes in EDA

H3	Low repayment ratio → risk	✓ Confirmed — REPAYMENT_RATIO strongly predictive (SHAP)
H4	Low LIMIT_BAL → risk	✓ Confirmed — negative LR coefficient; confirmed by SHAP
H5	Young age → higher risk (non-linear)	~ Partial — weak effect; non-linear pattern in EDA
H6	Class imbalance → use AUC/F1	✓ Confirmed — ~33% defaults; AUC used as primary metric
H7	Non-linear models > linear	✓ Confirmed — XGBoost AUC=0.721 > LR AUC=0.681

5. Business Impact

- Estimated savings vs no-model baseline: \$4,695,600
- Defaulters caught (TP): 1090 out of 1728 (63.1%)
- A client with $\text{PAY_0} \geq 2$ (2+ months overdue) is the highest-risk profile.
- REPAYMENT_RATIO below 0.10 is a critical early warning signal.
- Recommended production model: XGBoost (AUC = 0.721)